Forecasting Volatility: An Empirical Evaluation of Indian Stock Market

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Abstract

Forecasting volatility for stock market always remains a mystery for investors, speculators, banks, insurance, regulators and other financial institutions. Volatility, unlike other variables, cannot be observed from the market or true volatility can never be known. The existence of a plethora of forecasting volatility model, the selection of appropriate model becomes critical. Literature provides many alternatives for forecasting volatility for the stock market but consensus lacks on the selection of a specific model to forecast volatility. Along with "unconditional standard deviation", the most commonly use measure of volatility, present research has considered six others models of volatility forecasting, namely Standard GARCH (Generalized Autoregressive Conditional Heteroscedasticity) model, GJR-GARCH model, Exponential GARCH model (eGARCH), Asymmetric Power GARCH model (apGARCH), Component Standard GARCH model (csGARCH) , and Option Implied Volatility model to evaluate the best fit. To forecast volatility, the daily closing value of NSE Nifty 50 (Index) is considered from period 1st April 2002 to 31st March 2015, a span of 13 years. To remove in-sample biasness, present study has divided the data into in-sample data (estimation period), period from 1st April 2002 to 31st March 2009 and out-of-sample data (Backtesting Period), period from 1st April 2009 to 31st March 2015. It was found that no single model as the best performing model. The overall evaluation indicates that GJR-GARCH and option implied volatility are more consistent as compared to others. . Interestingly, most commonly used model i.e. unconditional standard deviation found to be the worst model among all models studied in present research. Further, the study highlights the type of volatility measure appropriate in different market scenarios.

Keywords: *Volatility, GARCH Family Models, Realized Volatility, Standard Deviation*

1 Introduction

The aftermath of Global financial crisis and Euro crisis, volatility held the attention of the academicians and practitioners. In general, volatility is the essence of the market. Without volatility market would just reduce to only a single instrument and monotonous investors. Just like life would not have any existence on the earth without diversity in nature, similarly financial market would not have any existence without volatility in the various financial products. Just imagine market without volatility. In such scenario, investors would only invest in those securities which offer higher returns. In this way, the entire investment would get restricted to only one security or the price of all securities adjusted till they provide a constant return. As no other security provide higher return, money would get invested till securities provide positive return and immediately return on all securities would reduce to zero. In such scenario, liquidity would get squeezed as there are no benefits of investing money. Further, entire derivative products would get disappeared as their only major purpose is to hedge against risk or speculate against unexpected movement which does not exist anymore. In short volatility is soul of the financial market without which financial market can not exist.

The debate on volatility in literature has not been settled yet. Volatility always remains a hot spot between researchers, academician, and practitioners. Still there is no consensus what constitute volatility and how it can be computed. Volatility, unlike other variables, cannot be observed from the market or true volatility can never be known. It can only be estimated from the past data. Standard deviation is most commonly use measure of volatility but it is symmetric. It considers both upside and downside variations in the stock price as a risk and treats equally. The definition of volatility vary from investor to investor and depends upon investor position in the market. The next section explain the most commonly used models for forecasting volatility.

2 Concept of Volatility and Forecasting Models

Volatility is a crucial as well as critical concept in theory and practice of finance. Nonetheless, it is not easy to define the concept of volatility. Since volatility is often calculated as a sample standard deviation, people often mistakenly assume that volatility is equivalent to standard deviation, which is in fact only a biased estimator of true volatility.

Volatility Forecasting Models

1. **Standard Deviation**

   Statistically, volatility refers to the sample standard deviation of the continuously compounded returns of a financial instrument with a specific time horizon. The sample standard deviation is defined as:

$$\tilde{\sigma} = \sqrt{\frac{1}{t}\sum_{i=1}^{t}[r_i - \bar{r}]^2} \tag{1}$$

   where, r_i is the return on i^{th} day , and $\bar{r}$ is the average return over the t-day period. Standard deviation is often used to quantify the risk of the instrument over that time period. The unconditional standard deviation assume that volatility is independent for each day. In reality, it is observed that volatility does not exhibit independent behaviour, instead, volatility appear in cluster. Abusing the terminology slightly, it could be stated that "volatility is autocorrelated". The behaviour in volatility when high volatility is likely to follow by high volatility and low volatility is likely to follow by low volatility is called volatility clustering or volatility persistence. The ARCH/GARCH family of models aim to capture volatility persistence or clustering.

2. **GARCH Family models**

   There are plethora of models in GARCH family models, but present study have considered only following models:

   (a) **The Standard GARCH Model**

   The generalized autoregressive conditional heteroscedasticity (GARCH) model was developed by Bollerslev (1986) . It may be written as:

$$\sigma_t^2 = \omega + \sum_{j=1}^{q}\alpha_j\epsilon_{t-j}^2 + \sum_{j=1}^{p}\beta_j\sigma_{t-j}^2 \tag{2}$$

   where, σ_t^2 denotes the conditional variance at time t, ω is the intercept or minimum variance of the stock, α_j represent the impact of information, arrived on j days before the t, on volatility at time t, β_j represent the impact of volatility, j days before the t, on volatility at time t, and ϵ_t^2 are the residuals (or difference of fitted variance and actual variance)

(b) **The GJR-GARCH Model**

The GJR GARCH model of Glosten, Jagannathan, and Runkle (1993) models positive and negative shocks on the conditional variance asymmetrically via the use of the indicator function I,

$$\sigma_t^2 = \omega + \sum_{j=1}^{q} (\alpha_j \epsilon_{t-j}^2 + \gamma_j I_{t-j} \epsilon_{t-j}^2) + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2 \tag{3}$$

where γ_j now represents the 'leverage' term. The indicator function I takes on value of 1 for $\epsilon \leq 0$ and 0 otherwise. So, the model uses threshold value as 0 to separate the impact of past shocks.

(c) **The exponential GARCH (EGARCH) model**

The exponential model of Nelson (1991) is defined as,

$$\log_e(\sigma_t^2) = \omega + \sum_{j=1}^{q} (\alpha_j z_{t-j} + \gamma_j (|z_{t-j}| - E|z_{tj}|)) + \sum_{j=1}^{p} \beta_j \log_e(\sigma_{t-j}^2) \tag{4}$$

where the coeficient α_j captures the sign effect and γ_j the size effect. Since, EGARCH model log of conditional variance ($\log \sigma_t^2$), so it does not require any restriction on the parameters of the model. So, the parameters of EGARCH model can take any value because log ensure that volatility always remain positive. Due to absence of any restriction on the parameters, EGARCH is very popular model among researchers.

(d) **The Component sGARCH model ('CGARCH')**

The model of Lee and Engle (1993) decomposes the conditional variance into a permanent and transitory component so as to investigate the long- and short-run movements of volatility affecting securities. Letting q_t represent the permanent component of the conditional variance, the component

GARCH model can then be written as:

$$
\begin{aligned}
\sigma_t^2 &= q_t + \sum_{j=1}^{q} \alpha_j(\epsilon_{t-j}^2 - q_{t-j}) + \sum_{j=1}^{p} \beta_j(\sigma_{t-j}^2 - q_{t-j}) \\
q_t &= \omega + \rho q_{t-1} + \phi(\epsilon_{t-1}^2 - \sigma_{t-1}^2)
\end{aligned}
\tag{5}
$$

where effectively the intercept of the GARCH model is now time-varying following first order autoregressive type dynamics. So, under component GARCH model, unconditional variance, which is function of $\{q_t\}$, is not constant. Component GARCH model can be mixed with any other models like GJR-GARCH, EGARCH or component GARCH by allowing the intercept term (ω) in the same way as defined in CGARCH model.

(e) **The asymmetric power ARCH model ('apARCH')**

The asymmetric power ARCH model of Ding, Granger, and Engle (1993) allows for both leverage and the Taylor effect. Taylor effect is named after Taylor (2012) who observed that the sample autocorrelation of absolute returns was usually larger than that of squared returns.

$$
\sigma_t^\delta = \omega + \sum_{j=1}^{q} \alpha_j(|\epsilon_{t-j}| - \gamma_j \epsilon_{t-j})^\delta + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^\delta \tag{6}
$$

where $\delta \in \Re^+$, being a Box-Cox transformation of σ_t, and γ_j the coefficient in the leverage term. The asymmetric power ARCH model ensure that fitted volatility $\{\sigma_t^\delta\}_{\delta \in \Re^+}$ is unbiased.

3. **Implied Volatility Models**

Implied volatility refers to the volatility at which Black-Scholes model (Black and Scholes, 1973) price equal to observed market price. Implied volatility represent expected volatility till the expiry of an option. It is believed that implied volatility contains information about future volatility, hence considered better measure of volatility in relative to various time series based model of volatility.

The most common model to derive the implied volatility is Black-Scholes (BS) model of European option price. BS model state that option price at time t is function of S_t (the price of underlying security), K (the strike price), r (the risk free interest rate), τ (time to option maturity) and σ (the volatility of the underlying assets over the period τ). Given the S_t, K, τ, r and observed market price, one can use a backward induction technique to derive a σ that is used as an input into the model. So, It can be said that implied volatility is implicitly contained in option price.

3 Literature Review

Akgiray (1989) was first one to test the predictive power of ARCH/GARCH models. Akigray investigated the daily returns on the CRSP value-weighted and equal-weighted indices, period from January 1963 to December 1986. He compared the performance of ARCH and GARCH, and the GARCH model was found to be more superior. Pagan and Schwert (1990) compared the performance among parametric and non-parametric models of volatility forecasting. Authors found supremacy of parametric models over non-parametric models in forecasting models. In parametric models, they found EGARCH as the best model because of its ability to capture the asymmetry effect in volatility. Study similar to Pagan and Schwert (1990) carried by Cao and Tsay (1992) , where, author compared the performance among ARMA(1,1), GARCH(1,1), EGARHC(1,1) and TAR model. Cao and Tsay found TAR provides the best forecast for large stocks and EGARCH gives the best forecast for small stocks, and they suspect that the latter might be due to a leverage effect. The study of Cumby, Figlewski, and Hasbrouck (1993) and Lee (1991) also found the dominance of EGARCH model over the historical models. Figlewski (1997) compared the performance between historical models and GARCH (1,1) model. Author found GARCH as best model for S&P 500 but ironically, he also found the same model as a worst model in the other market. Ederington and Guan (2000) forecasted the volatility using both absolute return and squared returns. They found that models which were based on absolute return deviations rather than models based on squared return deviation generally forecast better. Other studies likes Brailsford and Faff (1996) , Taylor (2004) found dominance of GJR-GARCH model while Frances and Franses and Dijk (1996) out rightly reject the use of GJR-GARCH model. Instead of superiority of single model, Ker and Clements (2008) found that the most accurate volatility forecast can be obtained from a combination of short and long memory models of realised volatility.

Literature provide evidence, wherein, authors did not found any significant difference in forecasting ability of various model or found historical models as best model to forecast volatility. For example, study by Brooks (1998) and McMillan, Speight, and Apgwilym (2000) found no significant improvement in forecasting ability of various models. Kumar (2006) and Sharma (2014) found Exponential Weighted Moving Average (EWMA) model as the best model to forecast market volatility.

Thenmozhi and Chandra (2013) compared the performance of India Volatility Index (India VIX) with traditional measures of stock price volatility, such as ARCH/GARCH models. They found that the India VIX was a better predictor of realised volatility than the GARCH and the EGARCH measures of conditional volatility and superiority of the India VIX for all the measures of realised volatility.

In spite of numerous literature on volatility, research are inconclusive. No single model perform best for all assets, under different time period and for different loss function. However, all the studies discussed above share one or more of the following characteristics: First, all studies test for large number of similar models designed to capture volatility persistence; second, they use a large number of error statistics which has a very different loss function; third, they used squared daily, weekly, monthly return as proxy of actual volatility, which result in extremely noisy volatilities estimate; and fourth, except very few, effects of external variable on the volatility is completely forgotten by the researchers.

4 Relevance of the study

Volatilities forecasting serve many purposes. When it is interpreted as uncertainty, it becomes a key input to many investment decisions and portfolio creations. Volatility is most important variable in the pricing of derivative securities. To price an option, we need to know the volatility of underlying securities from now until the option expires. Basel accord makes volatility forecasting a compulsory risk management exercise for many financial institutions around the world.

If volatility is forecasted more accurately, it would enable the investor to more accurately determine the premium for risk. Further, it would enable the regulator and risk managers to take more appropriate steps to manage risk. Research scholars who face a dilemma when forecasting volatility will also be able to choose a more appropriate model without going into the detailed investigation.

5 Objectives and Hypothesis

The present study aims to achieve following objectives:

1. To Identify and predict future volatility of Nifty from various models of volatility forecasting.

2. To compare the performance of various models of volatility forecasting with each other.

3. To compare the performance of various models of volatility forecasting with realized or actual market (Nifty) volatility.

So, based on the above objectives following hypothesis are formed:

- H01 : There is no difference in volatility forecasted by various models of volatility forecasting.

- H02 : The performance of all volatility models is symmetric.

6 Data Description

To forecast and evaluate the performance of volatility models, this study used secondary data for daily closing value of S&P CNX Nifty Index, daily data of Nifty Option contracts, daily closing value of India Volatility Index (VIX), Nifty future prices, and MIBOR. Details about the data given below in the Table 1: The data on CNX Nifty Index has been considered with the primary aim of forecasting volatility from various models. The data prior to 1st April 2002 is not considered because trading in Nifty future has introduced on June 4, 2001 and many empirical studies, for example, Raju and Karande (2003) , Bandivadekar and Ghosh (2003) and Singh and Kansal (2011) etc. have found significant decline in market volatility after the introduction of derivative product especially future and option contracts.

Table 1: Data Description

Series	Frequency	Beginning	End	Observations
Nifty	Daily	1st Apr 2002	31st Mar 2015	3242
VIX	Daily	1st Apr 2009	31st Mar 2015	1490
Nifty Option	Daily	1st Apr 2009	31st Mar 2015	23155
Nifty Future	Daily	1st Apr 2009	31st Mar 2015	-
MIBOR	Daily	1st Apr 2009	31st Mar 2015	1509

In case of option price, we have considered data for call option written on CNX Nifty Index during the period from 1st April 2009 to 31st March 2015. Since, Call-Put parity[1] ensures a definite bondage between call and put option prices therefore, data for the put option is not considered for this study. The data prior to 1st April has not been considered due to maintain similar data period with the VIX which is considered from 1st April 2009 to 31st March 2015. Future Price and MIBOR have been used as input into Black-Scholes model and hence considered in par with option price data.

7 Methodology

To achieve the objectives, present study considered seven models, namely Unconditional Standard Deviation (also written as Long Term Moving Volatility), Standard GARCH (Generalized Autoregressive Conditional Heteroscedasticity) model, GJR-GARCH model, Exponential GARCH model (eGARCH), Asymmetric Power GARCH model (apGARCH), Component Standard GARCH model (csGARCH) , and Option Implied Volatility model. To evaluate the performance of various volatility models, following steps were taken:

1. Volatility for NSE Nifty 50 from all models of volatility are forecasted, period from 1st April 2009 to 31st March 2015 (Backtesting Period) using in-sample-period (estimation period) of 1st April 2002 to 31st March 2009, on rolling basis. It involve re-estimating the parameters of the model as more recent data become available. The parameters of every GARCH-family models have been updated

[1] It is based on replicating portfolio and does not make any rigid assumption like by BS model.

after every 15 days [trading days] and for historical volatility models parameters are revised for each trading day. For example, first, model parameters are estimated using in-the-sample data, period from 1st April 2002 to 31st March 2009, then forecasting is made for next 15 trading days ie from 1st April 2009 to 22nd April 2009 (include only 15 trading days). After that in-the-sample period data is forwarded by 15 trading days from behind (now in-the-sample period span from 22nd April 2002 [2] to 22nd April 2009) and parameters are re-estimated including recent observations, ensuring that number of observations remain same every time in-the-sample period. Now, again forecasting will be made for next 15 trading days and procedure will be repeated till the entire out-of-sample period is covered. Similar procedure has been followed for every GARCH-Family model for forecasting volatility.

Present study comprises backtesting period period from 1st April 2009 to 31st March 2015, thus comprising total 1490 observations. If model is rolled after every 15 days, it means each model has been regressed exactly 100 times to forecast volatility for entire out of sample period. In our study, we have used five GARCH type models, thus, total 500 times GARCH models have been regressed. The purpose of the rolling volatility is to ensure that parameters are based on recent observations.

2. To construct benchmark volatility, square of daily return is considered as proxy of actual to evaluate the performance.

3. To find best performing model, four loss functions are being used along with Index of Rank Dominance. The four loss functions used are as follows:

   (a) Root Mean Square Error(RMSE)

$$RMSE = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(\hat{\sigma}_t - \sigma_t)^2} \tag{7}$$

[2]There are 15 trading days between 1st April 2002 to 21st April 2009

(b) Mean Absolute Error(MAE)

$$MAE = \frac{1}{N}\sum_{t=1}^{N} |\hat{\sigma}_t - \sigma_t| \tag{8}$$

(c) Mean Absolute Percent Error(MAPE)

$$MAPE = \frac{1}{N}\sum_{t=1}^{N} \frac{|\hat{\sigma}_t - \sigma_t|}{\sigma_t} \tag{9}$$

(d) CDC

$$\begin{aligned} CDC &= \frac{1}{N}\sum_{t=2}^{N} \zeta_t \\ \zeta_t &= \begin{cases} 1, \text{ if } (\sigma_t - \sigma_{t-1})(\hat{\sigma}_t - \hat{\sigma}_{t-1}) > 0 \\ 0, \; otherwise \end{cases} \end{aligned} \tag{10}$$

where σ_t is the realized volatility (or proxy for actual volatility) and $\hat{\sigma}_t$ is forecasted volatility. The higher the value of the statistics, worst is the performance of the model.

8 Analysis and Results

8.1 Descriptive Statistics of Forecasted Volatility

Descriptive statistics like mean, standard deviation, skewness, and kurtosis etc. provide a basic features of the forecasted volatility from all model. Descriptive statistics are also used to examine the consistency of various models. Descriptive statistics are shown in Table 2. The out-of-sample period start from 2-Apr-2009 because forecasted volatility for Implied Volatility model cannot be computed for 1-Apr-2009. To bring consistency in our summary statistics, we have removed first forecasted volatility of 1-Apr-2009 of all volatility models. The volatility from implied volatility model and VIX, first converted into daily volatility by assuming 248 trading days in year, as

$$DailyVolatility = AnnualVolatility/\sqrt{248}$$

Converting all volatility into daily volatility (otherwise, daily volatility can be con-

Table 2: Summary Statistics of Forecasted Volatility of all models

Particulars	sGARCH	gjrGARCH	eGARCH	csGARCH	apARCH	IV	LTMV	VIX
Observations	1489	1489	1489	1489	1489	1489	1489	1489
Start Date	02-Apr-09	02-Apr-09	02-Apr-09	02-Apr-09	02-Apr-09	02-Apr-09	02-Apr-09	02-Apr-09
End Date	31-Mar-15	31-Mar-15	31-Mar-15	31-Mar-15	31-Mar-15	31-Mar-15	31-Mar-15	31-Mar-15
Mean	0.012337	0.012264	0.012431	0.012064	0.01246	0.012644	0.017468	0.014012
Median	0.011177	0.01108	0.01131	0.010938	0.011207	0.011395	0.017538	0.012903
Maximum	0.066382	0.044028	0.03445	0.069395	0.050554	0.034173	0.018385	0.035604
Min	0.006596	0.006508	0.005622	0.006051	0.006275	0.006095	0.015432	0.007344
Std. Dev.	0.004908	0.004301	0.00424	0.004819	0.004281	0.004501	0.000701	0.004787
Skewness	3.88863	2.184133	1.197004	4.024351	1.80237	1.540243	-0.56644	1.545543
Kurtosis	27.87315	8.529318	2.054081	30.98296	7.756214	2.685149	-0.36967	2.964104
Annualized Mean	0.194283	0.193136	0.195771	0.189991	0.196217	0.199117	0.275091	0.220657

verted into annual volatility) ensure summary statistics of various models are comparable. The annualized mean volatility from all GARCH-Family model and Implied Volatility (IV) model is between 18% and 19%, whereas, same is 27% and 22% for Long Term Moving Volatility (LTMV) and Volatility Index (VIX) respectively. The high average from LTMV is due to high volatility during the 2008 and 2009(crisis period) which remained persistent until the end of estimation period. Except LTMV model, standard deviation of forecasted volatility from all the models is also a within a small range. Skewness of forecasted volatility from all model is positive except LTMV model. Kurtosis which represent probability mass in the tail of the distribution is highest for csGARCH model (30.98), followed by sGARCH model (27.873). Kurtosis for VIX and IV model is 2.96 and 2.68 respectively, which is very small in comparison to sGARCH and csGARCH model. High kurtosis of csGARCH and sGARCH is also apparent in maximum value of volatility forecasted. sGARCH forecasted a maximum volatility of 6.6%, whereas highest volatility from the VIX is 3.56% only.

Descriptive statistics shows that volatility forecasted from the various models have almost been consistent. LTMV model has been an exception among all the models. The variations in the descriptive statistics also ensure that volatility forecasted from the various models is not same. Some models produce extreme forecast more usual than others, whereas, some are more skewed than others. A quantitative measure of linear relationship is correlation. Correlation shows a degree of linear correlation between two variable. Correlation of 1 or -1 indicates that there is perfect positive or negative relationship between the two variables. Correlation matrix has been shown in Table 3.

Table 3: Correlation Matrix for volatility forecasted from various models

Model	sGARCH	gjrGARCH	eGARCH	csGARCH	apARCH	IV	LTMV	VIX
sGARCH	1							
gjrGARCH	0.952	1						
eGARCH	0.885	0.971	1					
csGARCH	0.997	0.949	0.884	1				
apARCH	0.899	0.97	0.988	0.902	1			
IV	0.732	0.737	0.733	0.739	0.712	1		
LTMV	0.283	0.317	0.363	0.293	0.348	0.327	1	
VIX	0.758	0.751	0.742	0.766	0.728	0.965	0.357	1

As Table 3 shows, volatility forecasted from sGARCH and csGARCH is almost same and correlation is more than 0.99. Similarly, correlation between volatility forecasted from eGARCH and apARCH is 0.988, almost linear relationship. The differences in the forecast emerge only for higher estimate of volatility. For example, for sGARCH and gjrGARCH model, higher estimate of forecasted volatility lie above the straight line (shown in second panel in first row), indicating sGARCH forecast volatility higher than gjrGARCH model during high volatility period. Overall, the volatility forecasted from all GARCH family models exhibit high degree of linear relationship among themselves, but they reject our first hypothesis that all volatility models forecast same volatility. The correlation between forecasted volatility from VIX and GARCH family models is also more than 0.72. We have also found high degree of correlation between VIX and Implied Volatility model, as both derived from option contracts data.

8.2 Graphical Comparison of forecasted models from actual volatility

All the volatility forecasted from all the models have been plotted with squared of daily return (realized volatility) and has been shown in Figure 1.

As shownn in Figure 1, realized volatility is very noisy and shows very frequent up and down in true volatility when squared of daily returns was used as a proxy for true volatility. Due to frequent jump in squared of daily returns, researcher prefer measure

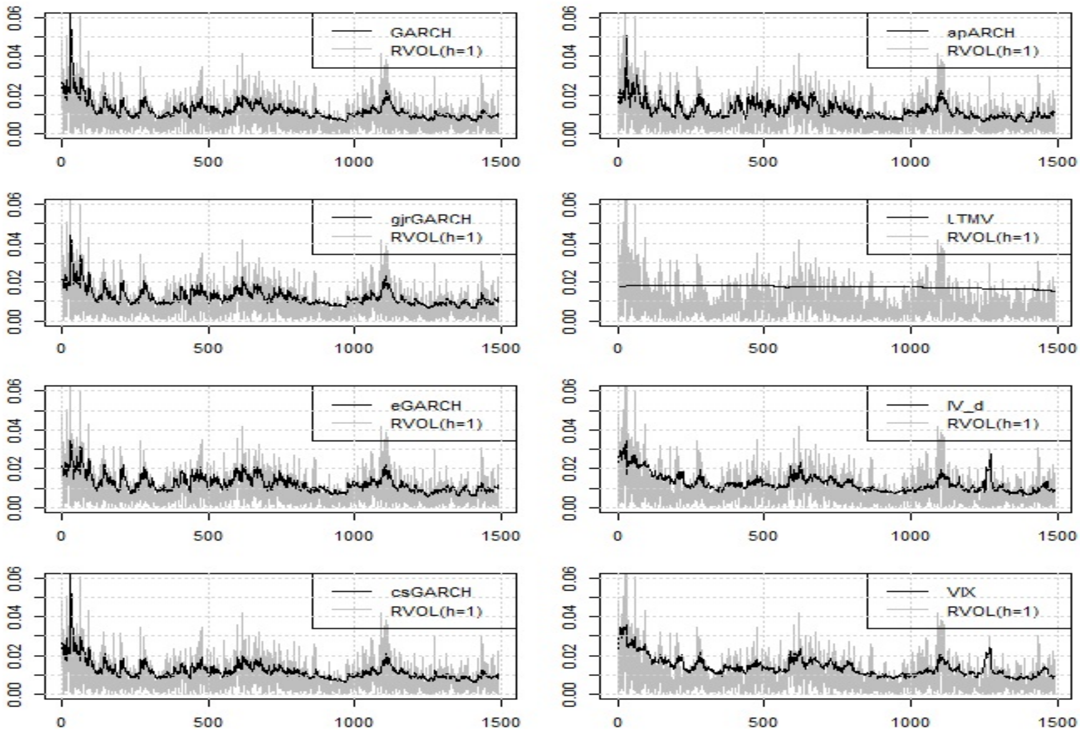

Figure 1: Graphical Comparison of Forecasted volatility with the Realized Volatility

which is sufficiently smooth. From the above graph, it is difficult to identify a single best model which best fitted to the true measure of volatility from the graphical comparison. Therefore, in the next section, all the models have been compared through sophisticated statistical techniques.

8.3 Statistical Comparison

The present study have used total four loss functions, namely Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percent Error (MAPE), and Correct Directional Change [3] with (CDC). The results of various volatility models from realized volatility is shown in Table 4.

Table 4 provide forecast error statistics from realized volatility. Table 4 exhibit clearly that performance of the model vary as loss function changed. Under the RMSE, Implied volatility has lowest value, whereas, gjrGARCH model is best performing model under MAE measure. Similarly, csGARCH and implied volatility (IV) model is best model under MAPE and CDC measures respectively. Under all measures except CDC, LTMV is the worst performing model. To get the overall performance of various

[3] CDC represents forecasting ability of model to produce right direction, therefore, higher is the value of Test Statistics, better is the rank .

Table 4: Forecast Error Statistics and Overall Ranking from Realized Volatility

	Error Statistics : RVOL(1)				RANK						
Model	RMSE	MAE	MAPE	CDC	RMSE	MAE	MAPE	CDC	Total Score	IRD	RIRD
sGARCH	0.009545	0.007061	7.772589	0.287828	6	5	3	6	20	0.7143	0.1786
gjrGARCH	0.009223	0.006928	7.753341	0.32616	3	1	2	5	11	0.3929	0.0982
eGARCH	0.009215	0.007	7.840736	0.33692	2	3	5	4	14	0.5000	0.1250
csGARCH	0.009438	0.006939	7.56935	0.28312	5	2	1	7	15	0.5357	0.1339
apARCH	0.00932	0.007035	7.896442	0.348352	4	4	6	3	17	0.6071	0.1518
LTMV	0.012316	0.010638	11.1611	0.357767	7	7	7	2	23	0.8214	0.2054
IV	0.009162	0.007097	7.78527	0.501681	1	6	4	1	12	0.4286	0.1071

model, we have computed total score and used Index of Rank Dominance (IRD). The best model is one that has lowest total score (because best model is ranked 1) or lowest value under IRD. We have found gjrGARCH as best performing model followed by Implied Volatility model. LTMV has found as the worst performing model.

9 Conclusion

Our results indicate that performance of various volatility models, when measured through various loss function with different measure of realized volatility, is highly critical to both measure of realized volatility and loss function. The choice of right model depends upon investor objective and kind of exposure in the financial market. If an investor is more concerned about the deviation of forecasting volatility from realized volatility, then the investor must use GJR-GARCH for forecasting volatility. If an investor is only concerned with extreme deviation in volatility and hedge against small variation then he must use option implied volatility. Similarly, if an investor is more concerned about direction of true volatility, then IV must be selected. The present study found standard deviation, the most commonly used measure of volatility, as worst measure of volatility forecasting.

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